# Portfolio Website PRD — Antigravity Build (Phase by Phase)

Optimized from your original prompt. Paste Section 1 into Antigravity once, then feed it one phase at a time.

## What I changed from your original prompt

- **The original says "build everything now."** For an IDE agent this produces a huge, shallow, buggy first pass. This version forces a phase gate: build, verify, report, stop.
- **Three animation libraries were unassigned.** Motion, GSAP and Lenis fight each other unless each has a clear job. Roles are now fixed (see Section 1).
- **Lenis and ScrollTrigger must share one clock.** The sync pattern is now spelled out, since this is the most common source of jitter.
- **App Router page transitions are limited.** Exit animations are unreliable in the App Router, so the spec asks for enter transitions plus a curtain overlay instead of a fragile `AnimatePresence` exit.
- **Vague quality lines became testable acceptance criteria** (Lighthouse, CLS, reduced motion, keyboard use, no TS errors).
- **Dependencies are capped.** No theme library, no extra UI kits. Custom text splitter, custom cursor, custom theme provider.
- **Content is separated from code** so you can swap in real details without touching components.
- **Placeholders are explicit** so the agent does not invent fake achievements.

Before you start, replace `[YOUR NAME]`, your initials, email and social links in Phase 0 when the agent asks, or edit `/data/site.ts` afterwards.

## Section 1 — Master context prompt (paste once, at the start)

```
You are a senior frontend engineer and motion designer. We are building a
premium, minimal, editorial developer portfolio. You will work PHASE BY
PHASE.

RULES OF ENGAGEMENT
1. Only work on the phase I give you. Never start the next phase on your own.
2. At the end of every phase: run `npx tsc --noEmit`, `npm run lint` and `npm
run build`. Fix every error. Then reply with: (a) what you built, (b) files
created or changed, (c) how to test it manually, (d) known issues. Then STOP
```

```
and wait for me to say "proceed to Phase N".
3. Make one git commit per phase: "phase N: <name>".
4. Do not add dependencies beyond the approved list. Ask first.
5. Do not invent personal facts, metrics or testimonials. Use clearly marked
placeholder content ("Project Name", "Your Role") in /data files.
6. If something in the spec is ambiguous or technically risky, pick the
simplest robust option, tell me in the report, and move on.

REFERENCE
https://tusharworks.vercel.app — use it only for creative direction, layout
philosophy, typography and hierarchy. Do not clone it. Original, more
refined, more minimal, better motion.

DESIGN DIRECTION
Ultra-minimal editorial. Typography is the main visual element. Generous
whitespace. Motion-rich but never overwhelming. Every animation has a
purpose. No glassmorphism, glows, loud gradients, generic cards or typical
AI-template look. Rounded corners on images are subtle (8–12px).

TECH STACK (approved list)
Next.js (App Router), TypeScript (strict), Tailwind CSS, motion (Motion for
React), gsap + @gsap/react (ScrollTrigger), lenis, lucide-react. Nothing else
without asking.

ANIMATION ROLES (do not mix them up)
- Lenis: smooth scrolling only. Single instance in a client provider. Drive
it from gsap.ticker (lenis.raf(time * 1000), gsap.ticker.lagSmoothing(0)) and
call ScrollTrigger.update on Lenis scroll, so there is exactly one animation
loop.
- GSAP + ScrollTrigger: scroll-linked work only (reveals tied to scroll
position, parallax, scrubbed timeline progress, section sequences). Always
use useGSAP with a scope so everything is cleaned up on unmount.
- Motion for React: component and UI-state animation (page-load sequence,
hover, springs, magnetic effect, cursor, menu open/close, theme toggle, route
enter transitions).
- Animate only transform and opacity. Never animate layout properties (width,
height, top, left, margin). Use will-change sparingly and remove it after
animation.

COLOR AND TYPE TOKENS (CSS variables, light default, dark via class)
Light: bg #F7F7F5, text #171717, muted #777777, border #E5E5E5.
Dark: bg #101010, text #F5F5F5, muted #999999, border #262626.
One subtle accent color, used sparingly (links, selection, status dot). Pick
one and define it as a token.
Fonts via next/font: Geist (body) and Space Grotesk (headings). Oversized,
tight letter spacing (-0.04em to -0.06em on display sizes), fluid sizes with
```

```
clamp().

ACCESSIBILITY AND PERFORMANCE BASELINE (applies to every phase)
- Respect prefers-reduced-motion: a useReducedMotion hook; when true, no
parallax, no scroll-jacking, no looping animation, Lenis disabled, content
visible immediately.
- On touch or coarse pointer devices: no custom cursor, no magnetic effects,
lighter animation.
- Semantic HTML, visible focus states, full keyboard support, skip-to-content
link.
- Text-split animations must keep the original text readable by screen
readers (aria-label on the parent, aria-hidden on split spans).
- No layout shift: reserve space for images (next/image with width/height or
fill + aspect ratio), fonts with display swap.
- Client components only where needed; keep server components for static
content.

PROJECT STRUCTURE
/app (layout.tsx, page.tsx, template.tsx, not-found.tsx, loading.tsx,
error.tsx, globals.css)
/components/ui, /components/motion, /components/layout, /components/sections
/data (site.ts, projects.ts, skills.ts, experience.ts)
/hooks, /lib, /public

Confirm you understand these rules, then wait for Phase 0.
```

## Section 2 — The phases

Each phase below is a separate prompt. Paste one, let the agent finish and report, test it yourself, then send the next.

### Phase 0 — Foundation

**Goal:** A clean, deployable skeleton with tokens, fonts, theme and content data.

**Prompt:**

```
PHASE 0: Foundation.
- Create the Next.js app (App Router, TypeScript strict, Tailwind, ESLint)
and the folder structure from the master prompt.
- Install only the approved dependencies.
- Define all color, spacing and type tokens as CSS variables in globals.css
and map them into the Tailwind theme. Light and dark.
- Load Geist and Space Grotesk with next/font. Create display, heading and
body utility classes with fluid clamp() sizes.
- Build a custom ThemeProvider (no theme library): class-based dark mode,
persisted in localStorage, respects system preference, and uses an inline
script in the head to prevent a flash of wrong theme.
- Create /data/site.ts (name, initials, role, email, location, timezone,
socials, status text, availability), /data/projects.ts (4 placeholder
projects with slug, title, year, description, stack, image, liveUrl,
githubUrl), /data/skills.ts (4 categories), /data/experience.ts (placeholder
entries across education, milestones, hackathons, open source, notable
projects). All typed.
- Global polish: custom ::selection color, subtle grain overlay (CSS or tiny
SVG, fixed, pointer-events none, very low opacity, togglable in one place),
smooth-scroll disabled for reduced motion.
- Root layout with full metadata skeleton (title template, description, Open
Graph, Twitter card, theme-color) driven from site.ts. Add favicon and a
placeholder OG image route.
- Page shows a simple placeholder so the app runs.
Acceptance: app builds, theme toggles without flash, no TS or lint errors,
all content comes from /data. Then report and STOP.
```

**Done when:** `npm run build` passes, hard refresh in dark mode shows no white flash.

## Phase 1 — Motion infrastructure

**Goal:** Reusable building blocks so later phases are fast and consistent.

**Prompt:**

```
PHASE 1: Motion infrastructure. Build and demo each piece on a temporary /dev page (delete or hide it before the report).
- SmoothScrollProvider: Lenis driven by gsap.ticker, synced with ScrollTrigger as defined in the master prompt. Disabled for reduced motion. Exposes a scrollTo helper via context.
- lib/motion.ts: shared easing curves, durations, spring presets, and stagger values. One source of truth.
- hooks: useReducedMotion, useIsTouch (coarse pointer), useMediaQuery, useMousePosition (rAF-throttled), useLocalTime(timezone).
- <SplitText> component: splits into lines/words/chars, handles resize re-split safely, accessible (aria-label + aria-hidden), with a mask-reveal variant (overflow hidden wrapper, text translates up from 100%).
- <Reveal>: scroll-triggered fade/translate reveal with stagger option, GSAP ScrollTrigger based, runs once.
- <MagneticButton>: Motion springs, pulls toward the pointer within a radius, resets on leave, no-op on touch and reduced motion, keyboard focusable with normal focus ring.
- <CustomCursor>: small dot plus follower, spring-based, states for default, link hover, and a text label (used later for VIEW PROJECT). Hidden on touch. Uses a context so any component can set the cursor state through data attributes or a hook.
- <AnimatedLink>: underline that draws in and out from the correct sides.
- <SectionHeader>: section number (01 / 02), animated horizontal rule drawing across, and title slot.
- <Preloader>: fast (about 1.2s max) initial reveal that hands off to the hero; skipped on reduced motion and on repeat visits in the same session.
Acceptance: demo page shows every component working; ScrollTrigger refresh works after route and font load; no memory leaks (cleanup verified); 60fps feel. Then report and STOP.
```

**Done when:** scrolling with Lenis is smooth, ScrollTrigger animations fire at the right positions, and nothing jitters.

## Phase 2 — Navigation

**Goal:** Floating nav with active section tracking and a polished mobile menu.

**Prompt:**

```
PHASE 2: Navigation.
- Floating minimal navbar: initials logo, Home, About, Projects, Experience, Contact, theme toggle.
- Entrance animation after the preloader. On scroll it subtly changes (reduced padding, background blur is NOT allowed; use a solid token background with a hairline border and slight scale or shrink). Hides on fast scroll down and shows on scroll up if it looks good, otherwise keep it fixed.
- Active section indicator via IntersectionObserver (not scroll listeners). Smooth scroll to sections through the Lenis helper, with correct offset.
- AnimatedLink underline on hover and focus.
- Animated theme switcher: icon morph between sun and moon using Motion; the page transition between themes is a short opacity/clip change, not a flash.
- Mobile: hamburger that animates to a close icon, full-screen menu with staggered masked link reveals, body scroll locked while open (stop Lenis), focus trapped, Escape closes, focus returns to the trigger.
- Include a skip-to-content link and proper nav landmarks, aria-current on the active link.
Acceptance: works with keyboard only; active link is correct on every section; no layout shift when the menu opens. Then report and STOP.
```

## Phase 3 — Hero

**Goal:** A fast, cinematic first impression.

**Prompt:**

```
PHASE 3: Hero.
- Full-screen (100svh) hero. Small intro: "HEY, I'M {name}". Massive
headline: "CREATIVE / DEVELOPER." (two lines). Supporting line: "I build
thoughtful digital experiences through code, creativity, and curiosity."
- Availability indicator (pulsing dot using opacity only, text from site.ts),
local-time indicator (useLocalTime), small "Currently building..." status
line from site.ts.
- Entrance: masked line reveal for the headline, staggered character motion,
supporting text fades up, other details follow. Total sequence under about
1.5s after the preloader. Staggering, never everything at once.
- Headline responds slightly to mouse movement (a few pixels of spring-based
translate, desktop only).
- MagneticButton for "Get in touch" (scrolls to contact).
- Animated scroll-down indicator that is a real button for keyboard users.
- Subtle background detail: very light animated lines or a fine dot grid
drifting with scroll. Must be tiny in performance cost; transform/opacity
only; static under reduced motion.
- Smooth transition out: as you scroll, hero content fades and moves slightly
(scrubbed ScrollTrigger) into the About section.
- Headline must be an h1 with correct accessible text. Fluid type that never
overflows at 320px.
Acceptance: no layout shift during the entrance; LCP element is text, not
blocked by animation (render text visible via CSS fallback if JS fails);
works at 320px to 2560px. Then report and STOP.
```

## Phase 4 — About

**Goal:** Editorial, asymmetrical, human.

**Prompt:**

```
PHASE 4: About.
- Section 01 with SectionHeader. Large heading "MORE THAN / JUST CODE."
- Asymmetrical two-column grid: expressive large paragraph on one side, smaller supporting paragraphs (background, interests, approach to building products) on the other. Short, meaningful placeholder copy in /data/site.ts, clearly marked to be edited.
- Minimal profile image placeholder (next/image, fixed aspect ratio, subtle rounded corners) with gentle scroll parallax (GSAP, small range, disabled for reduced motion and on mobile).
- Animated technology labels: small pill-free typographic labels that fade and shift in with a stagger and have a subtle hover shift.
- Scroll-triggered line reveals for the text via SplitText; animated divider drawing in.
- No skill progress bars.
Acceptance: reads well on mobile (single column, no parallax), text is real selectable text, no overlap at any breakpoint. Then report and STOP.
```

## Phase 5 — Selected work

**Goal:** The showpiece. Take the most care here.

**Prompt:**

```
PHASE 5: Selected Work.
- Section 02 with heading "SELECTED WORK." Render 4 projects from /data/projects.ts as large editorial rows, not cards. Alternate layouts for rhythm (image left/right, different aspect ratios, offset vertical alignment, oversized index numbers).
- Each project shows: name (animated title), year, short description, stack as typographic list, large preview image (next/image, rounded 8–12px, priority only for the first), Live demo link and GitHub link (real anchors with rel=noopener, open in new tab, clear aria-labels).
- Interactions: image scales slightly on hover (inner image scales inside an overflow-hidden frame), title animates on hover (letter or line shift), mouse-following "VIEW PROJECT" label via the CustomCursor label state (desktop only; on touch the whole row is simply tappable).
- On scroll: image reveal using a clip-path or mask wipe, subtle parallax inside the frame, staggered transitions between metadata items.
- The project row links to /projects/[slug] (page built in Phase 8; until then link target can be the live URL, and note this in the report).
- If a project has no image, use a tasteful typographic placeholder (not a broken image).
Acceptance: no layout shift on image load, hover states work with keyboard focus too (focus-visible equivalent), stable at 60fps while scrolling, graceful on mobile (stacked, no hover dependence). Then report and STOP.
```

## Phase 6 — Tools of the trade and Journey

**Goal:** Two typography-driven sections that stay calm and scannable.

**Prompt:**

```
PHASE 6: Skills and Experience.
SKILLS (Section 03): heading "TOOLS OF / THE TRADE." Clean typographic grid
with four categories: Frontend, Backend, AI & Machine Learning, Tools &
Workflow. Technology names reveal on scroll (staggered mask reveal), subtle
hover (shift or opacity change on the row, others dim slightly), thin
animated dividers between categories, optional interactive label that shows a
one-line note on hover or focus. No colorful logo cards.
EXPERIENCE (Section 04): minimal vertical timeline covering education,
milestones, hackathons, open source, notable projects, from
/data/experience.ts. The vertical line draws progressively with scroll
(scrubbed scaleY via ScrollTrigger), entries fade and slide in as the line
reaches them, the active entry is emphasized. Typography-focused: year,
title, place, one-line description. Include a type label per entry.
Both must be fully usable and readable with reduced motion and on mobile
(timeline collapses to a left-aligned single column).
Acceptance: data-driven (adding an item to /data updates the UI), no overflow
at 320px. Then report and STOP.
```

## Phase 7 — Contact and footer

**Goal:** Bold ending with obvious calls to action.

**Prompt:**

```
PHASE 7: Contact and Footer.
- Section 05 with huge headline "LET'S BUILD / SOMETHING." and supporting text "Have an idea, a project, or just want to connect? Let's talk."
- Oversized MagneticButton that opens a mailto link. Next to it, a copy-email button with clear feedback (text swaps to "Copied", icon morph, aria-live polite announcement, clipboard fallback if navigator.clipboard is unavailable).
- Links: Email, GitHub, LinkedIn, plus one or two other socials from site.ts, using AnimatedLink with hover text transformation (letters roll or slide).
- Smooth background transition as the section enters (page background eases to the inverse theme tone or a darker tone, using a scroll-linked opacity layer, not animated background-color on the whole body). Text contrast must remain AA in both themes.
- Cursor interaction: cursor grows over the big button.
- Footer: name, current year (computed), small back-to-top button with a smooth Lenis scroll and a tiny arrow animation, local time. Minimal.
Acceptance: every link works, copy feedback works, contrast verified, footer does not cause overflow. Then report and STOP.
```

## Phase 8 — Routes, transitions, states

**Goal:** Everything that makes it feel like one product rather than one long page.

**Prompt:**

```
PHASE 8: Routes, transitions and states.
- Build /projects/[slug]: a simple case study page generated statically with generateStaticParams from /data/projects.ts. Large title, meta (year, role, stack, links), hero image, short sections of placeholder copy, next-project link at the bottom. Include generateMetadata with per-project Open Graph data.
- Route transitions: because App Router exit animations are unreliable, implement app/template.tsx with a Motion enter transition (opacity + small y) plus a lightweight curtain overlay (transform only) triggered on link click that covers, navigates, then reveals. Scroll resets to top via Lenis after navigation. Must not break the browser back button. Disabled or instant under reduced motion.
- Custom 404 (not-found.tsx) matching the identity: huge "404" typography, short line, magnetic Back home button.
- loading.tsx and error.tsx (with a reset button) in the same visual style.
- Make sure ScrollTrigger refreshes after route changes and kills all triggers on unmount.
Acceptance: navigating home to a project and back works with both mouse and browser buttons, no duplicated listeners or stuck overlay, a deliberately broken route shows the 404. Then report and STOP.
```

### Phase 9 — Performance, accessibility, SEO, ship

**Goal:** Prove it is as good as it looks, then deploy.

**Prompt:**

```
PHASE 9: Hardening and deployment.
- Performance: run Lighthouse (mobile) on the production build. Targets:
Performance 90+, Accessibility 95+, Best Practices 95+, SEO 100. CLS under
0.05, no long tasks from animation setup. Fix what is below target: dynamic
import heavy client components, trim bundle size, check that GSAP plugins
registered once, verify images use correct sizes and formats, preload only
the fonts used above the fold.
- Throttle CPU 4x in DevTools and check scroll smoothness on the Projects
section; simplify parallax or disable it on low-power devices
(hardwareConcurrency and coarse pointer heuristics) if needed.
- Accessibility audit: keyboard-only walkthrough of every interaction, focus
order, focus visible everywhere, aria on custom controls, color contrast in
both themes, reduced-motion mode walkthrough (everything visible and usable
with no motion), screen reader check of split text.
- SEO: metadata complete on every route, dynamic OG image (next/og) with name
and role, sitemap.ts, robots.ts, JSON-LD Person schema, canonical URLs.
- Cleanup: remove the temporary /dev page and unused code and dependencies,
ensure `tsc --noEmit`, lint and build all pass with zero warnings.
- Deployment: add a README with setup, how to edit /data files, scripts, and
Vercel deployment steps; env variable template if any; verify the production
build runs with `npm run start`.
- Report Lighthouse scores before and after, and a short list of anything
intentionally left out.
Then report and STOP.
```

## Section 3 — Global acceptance checklist

Use this at the end to decide if it is really done.

- Lighthouse mobile: Performance 90+, Accessibility 95+, Best Practices 95+, SEO 100
- Zero TypeScript errors, zero lint errors, clean production build
- All content (name, links, projects, skills, experience, status) editable from `/data` only
- Reduced motion mode: fully readable and usable, no parallax, no scroll smoothing
- Mobile (320px to 430px): no horizontal scroll, no cursor effects, menu works, tap targets at least 44px
- Keyboard: every link, button, menu and toggle reachable with visible focus
- Light and dark both pass contrast checks; no theme flash on load
- No layout shift from fonts, images or animations

- Single animation loop (Lenis driven from GSAP ticker), no duplicate ScrollTriggers after route changes
- Every link and button works; copy-email shows feedback
- Custom 404, loading and error states match the visual identity

## Section 4 — How to run it in Antigravity

1. Open a new empty workspace and paste the Section 1 master prompt. Let it confirm.
2. Paste Phase 0. When the agent reports and stops, run `npm run dev` and check the acceptance line yourself.
3. If something is off, ask for fixes in the same phase ("Fix X, then re-run checks and report") before moving on. Do not stack the next phase on a broken one.
4. Only then say "proceed to Phase 1" and paste that phase's prompt.
5. Phases 1, 3 and 5 are the highest risk. Review them most closely, and test on a real phone before Phase 6.
6. If the agent starts to drift or the context gets long, start a fresh session, paste the master prompt again, and tell it "Phases 0 to N are complete; read the repo, then proceed with Phase N+1."

## Notes and tradeoffs

- **Blur is banned in the navbar** on purpose: backdrop blur on a fixed element is one of the most common causes of scroll jank on mid-range phones, and it also pushes the design toward the glassmorphism look you asked to avoid.
- **Custom SplitText instead of a plugin:** it keeps the dependency list short and lets you control accessibility. If you prefer GSAP's own SplitText, tell the agent in Phase 1.
- **Case study pages are the one addition beyond your original list.** They make route transitions and per-project SEO meaningful. If you want a single-page site only, delete Phase 8's first bullet and point project links to live demos.
- **Page transitions are enter-focused.** A true exit animation in the App Router needs workarounds that break easily with back/forward navigation; the curtain overlay gives a similar feel more reliably.